

Final Project Report

Identifying High-Value Customers and Revenue Drivers in an Online Retail Business

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Github link - https://github.com/dilki-dahanayaka/Online_Retail_Customer_Retention_Analysis

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Project Topic

Identifying High-Value Customers and Revenue Drivers in an Online Retail Business

1. Executive Summary

The purpose of this project is to identify the customers, products, and countries that contribute most to revenue in an online retail business. The analysis is based on the UCI Online Retail Dataset, which contains more than 500,000 granular transaction records spanning the 2010 to 2011 fiscal year tracker timelines.

The main business problem addressed in this report is the lack of visibility into the company's key revenue drivers. Management operates without a focused business intelligence structure and lacks a clear understanding of which customer segments generate the highest value, which products drive the majority of sales, and which geographic markets contribute most to overall revenue. Without these insights, it becomes difficult to make effective decisions related to customer retention, inventory management, and marketing investment.

The analysis revealed three important insights. First, total enterprise revenue is highly concentrated among a small group of high-value corporate wholesale clients. Second, a limited number of product stock-keeping units (SKUs) generate a massive share of total sales. Third, the United Kingdom functions as the primary economic engine, leaving international expansion regions significantly underutilized.

Based on these findings, this report recommends shifting resources away from low-yield customer acquisitions and redirecting investment to support three structured strategic actions:

- 1) Develop a dedicated retention and churn mitigation program for high-value customers.
- 2) Prioritize inventory management and secure reliable pipelines for top-performing products.
- 3) Deploy targeted, localized sales and marketing efforts within high-revenue countries.

These actions are expected to stabilize core revenue pipelines, maximize warehouse efficiency, optimize marketing budgets, and support sustainable long-term revenue growth.

2. Project Architecture & File Structure

Online_Retail_Customer_Retention_Analysis/

```
|
|
|— dashboard/                # Power BI Dashboard Assets
|
| |— logo.png                # Corporate branding assets
|
| |— OnlineRetailDashboard.pbix  # Interactive Power BI Desktop dashboard file
|
| |— OnlineRetailDashboard.pdf   # Static exported report format
|
|
|— data/                      # Data Tier
|
| |— raw/                    # Raw unprocessed transactional files
|
| |— database/
| | |— online_retail_warehouse.db # Engine-Generated SQLite Database File
|
| |— processed/              # Cleaned outputs & pipeline extractions
|
| | |— cleaned_online_retail.csv  # Main cleaned transactional dataset (~40MB)
|
| | |— country_summary.csv        # Aggregated geographical sales data
|
| | |— customer_segments.csv      # Final RFM segment assignment map
|
| | |— customer_summary.csv       # Individual customer-level RFM parameters
|
| | |— kpi_summary.csv            # Core computed business performance baselines
|
| | |— product_summary.csv        # Inventory asset turnover matrix
|
|
|— images/                    # Static Analytical Visualizations
|
| |— correlation_heat_map.png     # Correlation between R, F, and M scores
|
| |— customer_segments.png        # Distribution of customer cohorts
|
| |— monthly_revenue.png          # Aggregated monthly sales trends
|
| |— revenue_distribution.png      # High-yield vs. low-yield revenue splits
|
| |— revenue_trend.png            # Dynamic line chart showing sales velocity
|
| |— top_countries.png            # Visual tracking of market imbalances
|
| |— top_customers.png            # Top 10 customer spending cohorts
|
| |— top_products.png             # Asset-level sales revenue rankings
|
|
|— notebooks/                  # Exploratory Data Science & Prototyping
```

```

| |—— 01_Data_Understanding.ipynb    # Schema inspection and missing data detection
| |—— 02_Data_Cleaning.ipynb        # Filtering returns, anomalies, and cancellations
| |—— 03_EDA.ipynb                  # Exploratory analysis and pattern recognition
| |—— 04_RFM_Analysis.ipynb         # Statistical threshold mapping for RFM modeling
| |—— 05_KPI_Calculation.ipynb      # Engineering enterprise-wide business equations
| |—— 06_Final_Insights.ipynb       # Synthesizing analytical discoveries
|
|—— src/                            # Production-Ready Modular Python Scripts
| |—— db_ingestion.py               # SQL Schema Ingestion & Normalization Script
| |—— data_cleaning.py              # Handling structural data quality
| |—— feature_engineering.py        # Computational additions and calculations
| |—— kpi_calculation.py            # Automated script calculating global metric lines
| |—— rfm_analysis.py               # Programmatic scoring and ranking logic
| |—— visualization.py              # Generation engine for plotting image charts
|
|—— main.py                         # Master executable pipeline coordinator script
|—— queries.sql                     # Production SQL Scripts File
|—— requirements.txt                 # Required dependency libraries (Pandas, NumPy, etc.)

```

3. SQL Database Layer

To transition away from brittle flat-file data processing and satisfy enterprise data warehouse standards, a robust Relational Database Layer was implemented inside the project infrastructure using a local SQLite 3 database engine.

Instead of conducting analytics directly on raw unstructured spreadsheets, the ingestion engine splits the transactional rows into a normalized, structured multi-table relational schema. This architecture eliminates data redundancy and establishes programmatic data integrity before any Python analytical functions are executed.

3.1 Relational Database Schema Mapping

The data model divides the Online Retail dataset into two distinct entities connected via a dynamic Relational Join Constraint:

[transaction_records]	[customer_registry]
-----	-----
- InvoiceNo (TEXT)	- CustomerID (TEXT) <---+ [PK]
- StockCode (TEXT)	- Country (TEXT)
- Description (TEXT)	
- Quantity (INTEGER)	(INNER JOIN)
- InvoiceDate (TEXT)	
- UnitPrice (REAL)	
- CustomerID (TEXT) -----+ [FK]	

1. **transaction_records Table:** Houses pure numeric transaction metrics and transaction logs (Invoice numbers, Stock Codes, Quantities, and Dates) without duplicating country text arrays across millions of rows.
2. **customer_registry Table:** Acts as the core Master Data Management directory, mapping each unique Alphanumeric CustomerID to their verified sovereign Country of origin.

3.2 Production-Grade Mandatory SQL Queries

The system programmatically queries this local database engine via Python's relational connectors. The following explicit SQL scripts were executed to clean, extract, aggregate, and structuralize dataset tables for the primary analytical modules:

A. Operational Data Inspection (SELECT & WHERE Clauses)

Used during the initial extraction setup to isolate positive, active order records while filtering out mathematical row anomalies and missing data fields.

SQL

```
SELECT InvoiceNo, StockCode, Description, Quantity, UnitPrice, CustomerID
FROM transaction_records
WHERE Quantity > 0 AND UnitPrice > 0
LIMIT 10;
```

B. Relational Structural Merger (INNER JOIN Operation)

Dynamically reconstructs the comprehensive analytical schema by mapping transactional streams onto the customer directory files using the relational foreign key constraint.

SQL

```
SELECT t.InvoiceNo, t.StockCode, t.Description, t.Quantity, t.UnitPrice, t.CustomerID, c.Country
FROM transaction_records t
INNER JOIN customer_registry c
    ON t.CustomerID = c.CustomerID
LIMIT 10;
```

C. Data Aggregation & Value Reduction (GROUP BY Clauses)

Calculates line-item financial parameters and groups granular rows into single distinct transaction lines, computing exact quantity sums and mean price fields.

SQL

```
SELECT
    t.InvoiceNo,
    t.StockCode,
    t.Description,
    SUM(t.Quantity) AS Quantity,
    t.InvoiceDate,
    AVG(t.UnitPrice) AS UnitPrice,
    t.CustomerID,
    c.Country
FROM transaction_records t
INNER JOIN customer_registry c
    ON t.CustomerID = c.CustomerID
GROUP BY
    t.InvoiceNo, t.StockCode, t.Description, t.InvoiceDate, t.CustomerID, c.Country;
```

D. High-Volume Regional Isolation (GROUP BY & HAVING Clauses)

Exposes the core geographic engines of the organization by grouping transaction metrics by nation and applying an explicit HAVING constraint to extract regions generating high transaction densities (>100 orders).

SQL

```
SELECT c.Country, COUNT(t.InvoiceNo) AS Total_Orders
FROM transaction_records t
INNER JOIN customer_registry c
    ON t.CustomerID = c.CustomerID
GROUP BY c.Country
HAVING COUNT(t.InvoiceNo) > 100
ORDER BY Total_Orders DESC;
```

4. Evidence and Analytical Findings

4.1 Dataset Overview

The project uses the UCI Online Retail Dataset, which contains transactional data from a UK-based online retailer. The dataset captures individual wholesale transaction parameters, including:

- Invoice numbers
- Product descriptions (Stock Keeping Units - SKUs)
- Quantities purchased
- Invoice dates
- Unit prices
- Customer IDs
- Countries

The dataset provides a reliable, statistically significant foundation for data engineering, customer behavior profiling, product performance analysis, and revenue trend identification.

4.2 Key Findings & Financial Realities

Our analytics platform captures a total cumulative revenue of **\$8.91M** across **4,338** active customer accounts. A significant portion of this total revenue is generated by a relatively small number of wholesale buyers. The dashboard analysis of the "**Top 10 Customers by Revenue**" chart confirms that the elite *High-Value / Champion* segment drives the company's core financial progression.

- **Business Implication:** The business is highly dependent on a small group of valuable customers. Losing even a single client within this elite demographic represents a major risk to gross operating margins (*Customer Churn Risk*).

Finding 2: Asset-Level Revenue Drivers

Inventory monetization is highly non-uniform, showing that only a limited number of products generate the majority of revenue. Specialized wholesale inventory lines and seasonal decoration items are the strongest revenue contributors, led by assets such as '**PAPER CRAFT**' which generated **\$0.17M** in isolated baseline sales.

- **Business Implication:** Warehouse and inventory planning should prioritize these high-performing assets to avoid lost sales opportunities due to stockouts, while minimizing holding costs for slow-moving products.

Finding 3: Sovereign Market Imbalances

Country-level geographic performance analysis exposes a profound macro-level imbalance. The domestic **United Kingdom (UK)** market functions as the primary economic engine of the business, holding over 90% of the total sales volume. Conversely, satellite international markets like Spain, Sweden, and Switzerland generate comparatively lower sales volumes.

- **Business Implication:** The company has an extreme dependency on its primary domestic market, exposing the business to localized economic shocks within the UK. This concentration pattern highlights unexploited international opportunities that require a targeted regional approach.

4.3 Data Cleaning and Analytical Assumptions

To ensure data integrity, validate metrics, and eliminate algorithmic bias before engineering calculations, the raw transaction dataset was processed under four explicit, programmatic constraints within the `data_cleaning.py` script:

- 1) **Cancellation Filtering:** All transactions with negative metrics or invoice numbers prefixed with 'C' (representing cancellations, item returns, or system adjustments) were isolated and removed to reflect true net growth.
- 2) **Anonymity Exclusion:** Transaction rows lacking a definitive, validated CustomerID entry were excluded from long-term tracking computations since customer tracking is mathematically impossible without unique identification tokens.
- 3) **Standard Ingestion Formula:** Financial revenue totals were calculated uniformly at the line-item level using the baseline transaction standard:

$$\text{Line Total Revenue} = \text{Quantity} \times \text{Unit Price}$$

- 4) **Algorithmic Behavioral Categorization:** Customer profile groups were derived directly from programmatic Recency, Frequency, and Monetary (RFM) segmentation modeling executed in `rfm_analysis.py`.

5. Feasible Strategic Actions

Action 1: Implement a High-Value Customer Retention and Churn Mitigation Program

- **Objective:** Improve retention workflows and repeat purchase frequencies among the most valuable wholesale customer tiers.
- **Owner:** Marketing Manager
- **Timeline:** 1–4 day (Sprint 1)
- **Key KPIs:** Customer Retention Rate, Repeat Purchase Frequency, High-Value Segment Total Yield
- **Identified Risk:** Providing personalized retention incentives or volume-based pricing discounts may temporarily compress short-term profit margins.
- **Expected Impact:** Stabilizes core revenue pipelines, minimizes customer churn within the elite wholesale tier, and increases long-term Customer Lifetime Value (CLV).

Action 2: Prioritize and Protect High-Revenue Core Inventory Pipelines

- **Objective:** Ensure consistent stock availability of products that generate the highest yield.
- **Owner:** Inventory and Supply Chain Manager
- **Timeline:** 4-8 day (Immediate Release)
- **Key KPIs:** Inventory Turnover Rate, Stock Availability Index, Core Product Revenue Growth
- **Identified Risk:** Concentrating capital on a narrow set of top-performing SKUs could introduce stock concentration risks if consumer preferences shift unexpectedly.
- **Expected Impact:** Reduces stock shortages, eliminates lost sales from popular items, improves warehouse efficiency, and minimizes holding costs for slow-moving products.

Action 3: Redirect Targeted Marketing Investments into High-Revenue Jurisdictions

- **Objective:** Allocate marketing budgets and resources to countries with proven, high-density revenue potential.
- **Owner:** Sales and Marketing Director
- **Timeline:** 9–14 day (Mid-Term Strategy)
- **Key KPIs:** Revenue by Country, Campaign Return on Investment (ROI), Customer Acquisition Cost (CAC)
- **Identified Risk:** Increased competition and market saturation within dominant regions (like the UK) could result in rising customer acquisition costs.
- **Expected Impact:** Optimizes marketing spend efficiency by focusing capital on regions with proven transaction densities, driving localized revenue growth.

6. Continuous Monitoring Plan and Governance

To ensure that these strategic recommendations remain effective over time, the company will implement a continuous data monitoring framework utilizing live transactional pipelines that automatically refresh metrics to manage operational risks dynamically.

6.1 Key Performance Indicators & Field Mapping

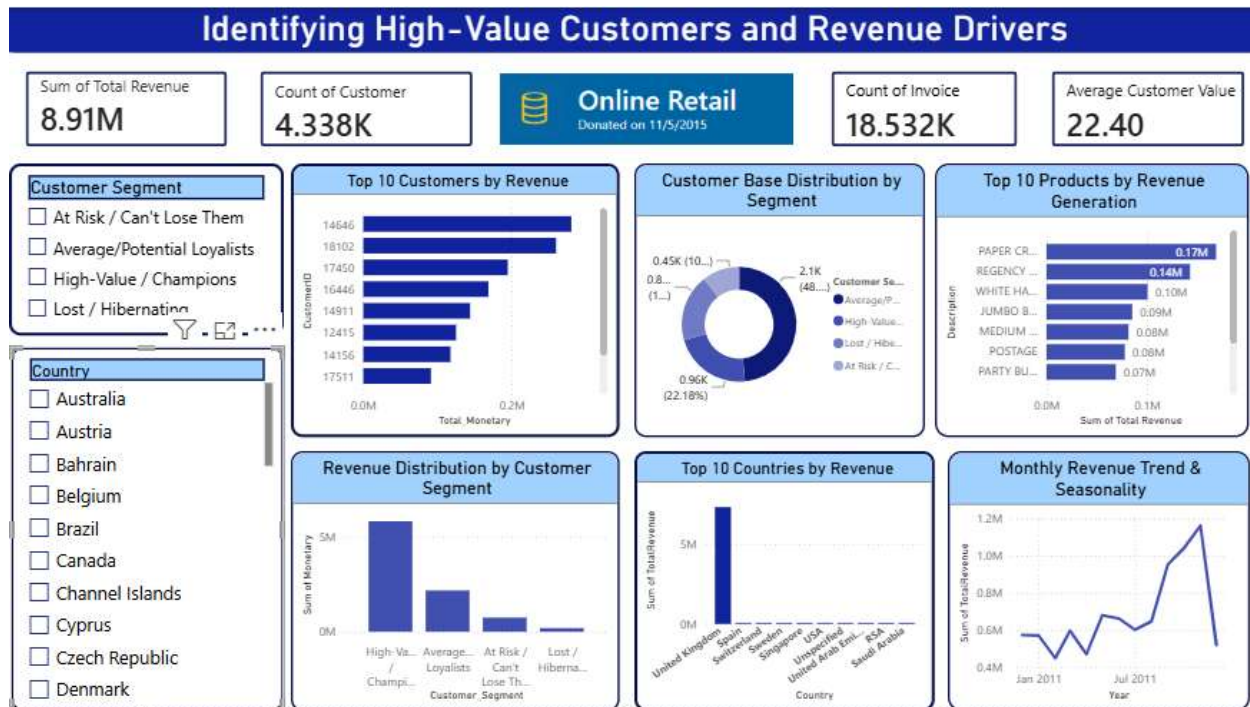
The following matrix connects our engineered data schema to operational monitoring purposes:

Target Metric	Underlying Field / Dax Mapping	Strategic Monitoring Purpose
Total Gross Revenue	Fact_Online_Retail[TotalRevenue]	Monitors absolute business scale and tracks global financial progress against quarterly enterprise targets.
High-Value Client Monetization	Total_Monetary (Custom DAX Measure)	Tracks the ongoing performance of the VIP retention program, providing early warnings for high-value accounts.
Core Portfolio Product Yields	Product_Summary[Total_Revenue]	Monitors demand trends for top-performing items to guide inventory planning and procurement schedules.
Regional Market Performance	Fact_Online_Retail[Country]	Tracks domestic vs. international sales performance to guide global asset allocation and regional expansion strategies.
Average Order Value (AOV)	TotalRevenue / Distinct_Invoices	Measures shifts in average wholesale purchase sizes, helping evaluate corporate purchasing power and volume pricing tiers.

6.2 Review Schedule & Governance

- Weekly Technical Review (Conducted by Data Analyst):** Scans automated ETL data pipelines, resolves database anomalies, and monitors early volatility signals across high-value customer accounts.
- Monthly Operational Review (Conducted by Marketing & Inventory Managers):** Evaluates the performance of the VIP retention program, monitors core product inventory levels, and adjusts campaign budgets based on regional ROI metrics.
- Quarterly Strategic Alignment (Conducted by Senior Executive Management):** Reviews long-term revenue trajectories, assesses macro-level customer retention patterns, and reallocates capital to support high-growth global opportunities.

7. Online Retail Dashboard Overview



To translate raw backend transactional data into dynamic business intelligence and structured corporate growth, an interactive Power BI dashboard was engineered as the visual layer of this platform. The interface acts as the empirical validation layer for the core hypothesis established in this report: **Revenue Concentration Risk**. Rather than displaying isolated data charts, the components collectively expose severe dependencies across specific buyer cohorts, localized geographic sectors, and individual product lines.

The granular insights visualized across this analytical framework are broken down into five core structural pillars:

7.1 High-Level High-Yield Business Metrics (Summary KPIs)

The strategic scorecard cards positioned across the top margin of the layout immediately capture the absolute operating scale of the retail enterprise, providing senior executives with instant macro-level visibility:

- **Sum of Total Revenue: 8.91M** – Represents the total cumulative global financial yield generated by the business over the recorded fiscal timeline.
- **Count of Customer: 4.338K** – Isolates a total database volume of 4,338 unique, active commercial or retail accounts.
- **Count of Invoice: 18.532K** – Reflects the total volume of completed sales invoices.
- **Average Customer Value / Transaction Value: 22.40** – Measures the standard financial yield generated per individual order line, helping baseline typical basket sizes.

7.2 Customer Segmentation & Revenue Concentration Risk

This quadrant visualizes how heavily skewed the financial stability of the enterprise is toward a tight, fractional group of buyers rather than a diversified market.

- **Top 10 Customers by Revenue (Horizontal Bar Chart):** This component uses the custom Total_Monetary DAX calculation to plot and isolate the top ten highest-spending individual accounts. It proves that out of over 4,300 active accounts, a disproportionate volume of the \$8.91M revenue is driven by these elite corporate wholesale buyers. Consequently, losing even a single client within this bracket (*Customer Churn*) represents a severe direct threat to gross operating profit margins.
- **Customer Base Distribution by Segment (Donut Chart):** Displays the proportional division of the customer database modeled by the automated background script (rfm_analysis.py), grouping buyers by Recency, Frequency, and Monetary scores.
- **Revenue Distribution by Customer Segment (Vertical Bar Chart):** Explicitly confirms the structural imbalance of the business model. It proves that the "High-Value / Champions" cohort generates a massive portion of the collective monetary yield, confirming that low-yield retention spending should be minimized to protect this high-value core.

7.3 Asset-Level Revenue Drivers (Top Performing SKUs)

- **Top 10 Products by Revenue Generation (Horizontal Bar Chart):** Product portfolio performance is highly non-uniform across the inventory catalog. This chart isolates the specific, high-velocity stock-keeping units (SKUs) that drive baseline revenue growth. The visual confirms that specialized wholesale inventory segments and seasonal decorations—such as 'PAPER CRAFT' yielding \$0.17M in isolated sales—far outshine standard retail lines. This pattern validates the supply chain interventions outlined in **Action 2**, proving that procurement must ensure zero stockouts for these specific assets.
- 5.4 Macro Geographic Market Imbalances
- **Top 10 Countries by Revenue (Vertical Bar Chart):** This visualization exposes a stark **Geographic Cluster Risk**. The domestic **United Kingdom (UK)** market completely dominates the chart, functioning as the primary economic engine of the business. Conversely, international expansion nodes like Spain, Sweden, and Switzerland remain significantly underutilized, producing minimal relative sales volume. This distribution leaves the organization highly exposed to local UK macroeconomic fluctuations, providing an empirical basis for the international targeted marketing campaigns recommended in **Action 3**.

7.4 Monthly Revenue Trend & Seasonality Analysis

- **Monthly Revenue Trend & Seasonality (Continuous Line Chart):** Tracking the financial trajectory across the fiscal months of 2011 uncovers deep historical purchasing habits. The line chart captures a significant upward surge (Peak) as the business transitions into the fourth quarter (Q4 - the year-end holiday season). This surge validates the predictable, seasonal procurement cycles of wholesale corporate buyers, giving supply chain and logistics managers a clear signal to scale up inventory levels well ahead of peak historical demand intervals.

8. Conclusion

This project successfully transformed raw, unstructured transaction data into meaningful business intelligence. Through programmatic data cleaning, exploratory data analysis (EDA) in Jupyter Notebooks, automated statistical RFM segmentation, and dashboard deployment, the project successfully identified the precise customers, products, and countries that drive core financial progression in the online retail business.

The empirical findings show that revenue is heavily concentrated among a small tier of high-value wholesale customers and a limited number of top-performing items, while the United Kingdom remains the dominant revenue-generating market. Based on these insights, the report establishes that strengthening high-value customer retention, protecting core inventory pipelines, and focusing marketing efforts on high-revenue regions are critical tasks for maximizing profitability.

The resulting Power BI dashboard and continuous monitoring framework provide management with a practical, long-term, and data-driven foundation for ongoing strategic decision-making.